Assignment 0: Build Your LC-3 Survival Kit

Overview

In this assignment, you will design and assemble your own **LC-3 Survival Kit**—a personalized set of study and problem-solving tools to help you master the LC-3 assembly language and computer architecture concepts.

This kit should be practical, easy to use, and tailored for students like you, who will rely on it throughout the course.

You will use the provided reference documents (ISA appendix, LC-3 manual, and lab examples) as your **core material**, but the final design and layout of the kit is up to you. Creativity and accessibility will be rewarded!

Required Contents

Instruction Reference Cards

Create 15 double-sided cards, one for each LC-3 instruction. Front: concise definition, opcode structure, and effects. Back: one worked example in LC-3 assembly.

Instruction Structure Table

A compact, one-page table listing the binary format of all LC-3 instructions, with fields labeled (DR, SR1, imm5, etc.).

Conversion Table

Quick reference for converting between binary, decimal, and hexadecimal. Include ASCII encodings for characters to aid in I/O instructions (e.g., PUTS, GETC).

Memory System Map

Visual diagram of the LC-3 memory system (Trap Vector Table, Interrupt Vector Table, OS/Supervisor stack, User program space, Device registers). Include registers (R0–R7, PC, IR, PSR, Condition Codes).

Main Working Sheet

Four columns: Address | Hex (4-digit) | Binary (16-bit) | LC-3 Assembly. This will be your active coding scratchpad.

Register State Tracker

Scratch paper template to track register changes step-by-step during program execution.

Deliverables

- A **compiled kit** (digital or physical).
- Organized, clean, and ready for immediate use in problem solving and labs.
- Optional: creative touches such as color coding, laminated sheets, or foldable guides.

Evaluation Criteria (100 points)

Category	Description	Points
Completeness	All required six components are included.	25
Accuracy	Technical correctness of instruction definitions, examples, tables, and encodings.	25
Clarity & Accessibility	Easy to navigate, visually clear, helpful for quick reference in class/labs.	25
Creativity & Usefulness	Layout/design enhances learning (color coding, grouping, visual aids, compactness).	25

Bonus (+5 points): Include a short reflection (½ page) on how you would personally use this kit to debug or write LC-3 programs.

Submission

- Bring your completed Survival Kit to class from **15th September**.
- Submit a **PDF scan** (or digital version) to the course portal for evaluation before the class time on **22nd September, 11:50 A.M. (Madison, NJ Time)**.